

Classic Deep CNN Architectures

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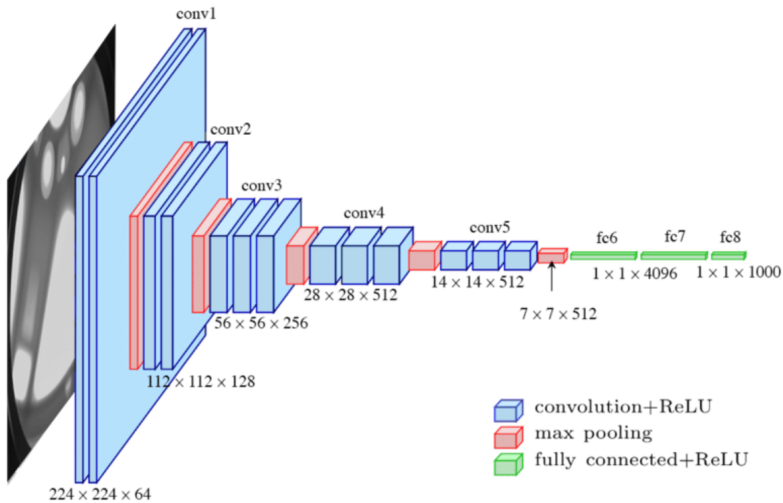
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By the end of this session, you will be able to:

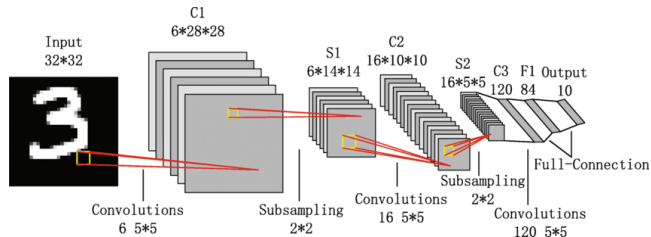
- ▶ See how deep CNNs have evolved over time.
- ▶ Get to know influential architectures: LeNet, AlexNet, VGG, InceptionNet, ResNet, EfficientNet, and MobileNet.
- ▶ Understand the key innovation behind each model (e.g., ReLU + dropout, small stacked filters, inception modules, skip connections, compound scaling, depthwise separable convolutions).
- ▶ Connect these architectures to the ImageNet benchmark that drove their development.
- ▶ Spot the limitations of CNNs and think about what's next.

CNN Architectures

Over time, researchers built advanced CNN architectures to improve performance and efficiency. These architectures introduced key innovations:

- ▶ **LeNet [LeCun et al., 1998]**: The first CNN architecture, designed for handwritten digit recognition.
- ▶ **AlexNet [Krizhevsky et al. 2012]**: The first CNN to achieve breakthrough performance on image classification.
- ▶ **VGGNet [Simonyan and Zisserman, 2014]**: Used very deep networks (up to 19 layers).
- ▶ **InceptionNet (GoogLeNet) [Szegedy et al., 2014]**: Used multiple filter sizes per layer (Inception modules).
- ▶ **ResNet [He et al., 2015]**: Introduced skip connections for training very deep networks.
- ▶ **EfficientNet [Tan and Le, 2019]**: Found a scaling method that simultaneously scales a CNN's depth, width, and resolution optimally using a single scaling coefficient.
- ▶ **MobileNet [Howard et al., 2017]**: Designed for mobile and embedded vision applications, using depthwise separable convolutions.

CNN Architectures: **LeNet-5**

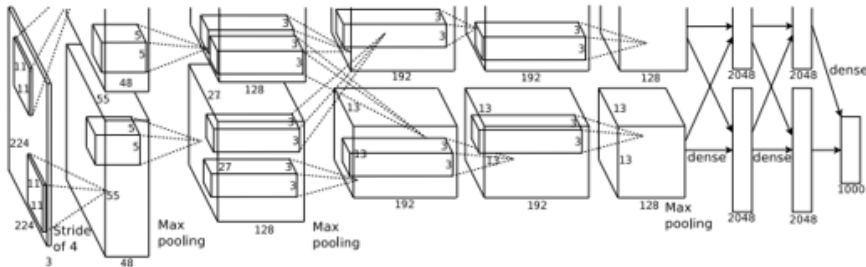


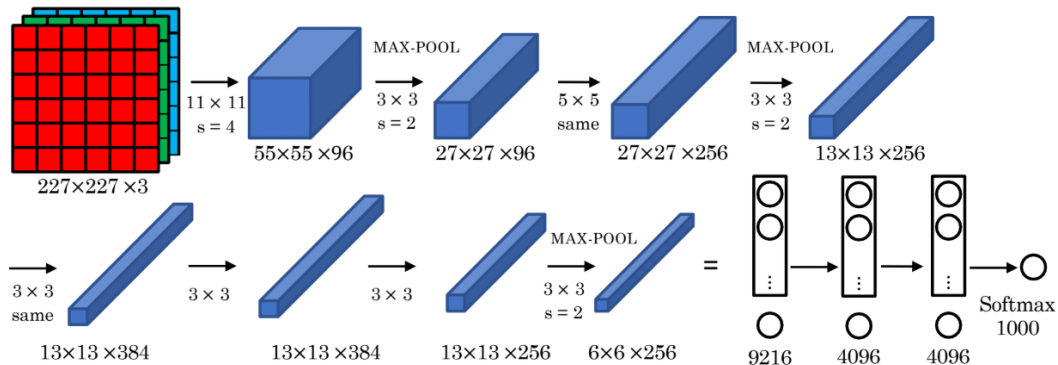
LeNet-5 architecture.

- ▶ LeNet-5 is a pioneering convolutional neural network (CNN) architecture developed by Yann LeCun and his collaborators in the late 1980s and early 1990s.
- ▶ It was built to recognize handwritten digits, like those in the MNIST dataset.
- ▶ The network uses layers that look for patterns (convolution), shrink images (pooling), and finally make decisions (fully connected layers).
- ▶ LeNet-5 introduced ideas like using filters to find features and pooling to reduce size, which are now common in modern CNNs.

CNN Architectures: **AlexNet**

- ▶ First big improvement in image classification.
- ▶ Made use of CNN, pooling, dropout, ReLU and training on GPUs.
- ▶ 5 convolutional layers, followed by max-pooling layers; with three fully connected layers at the end

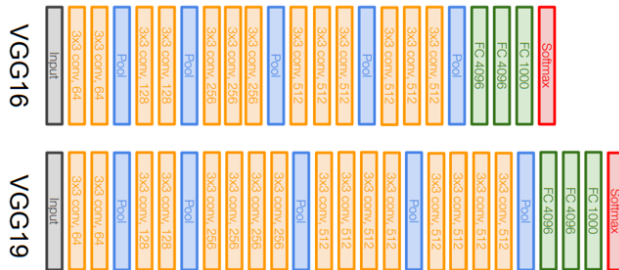




[Krizhevsky et al., 2012. ImageNet classification with deep convolutional neural networks]

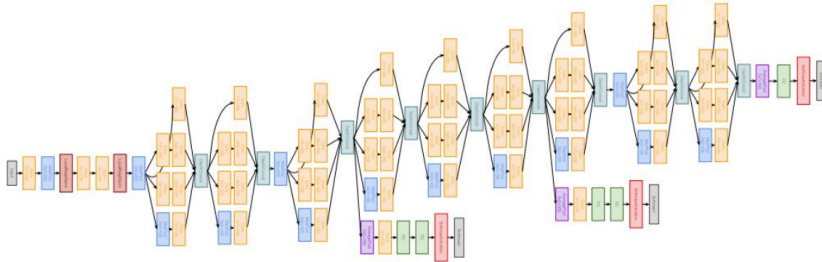
CNN Architectures: **VGGNet**

- **Improvement over AlexNet:** Uses a deeper network with small filters instead of a shallow network with larger filters.
- A stack of 3x3 conv layers (vs. 11x11 conv) has same receptive field, more non-linearities, fewer parameters and deeper network representation.
- Two variants: VGG16 or VGG19 conv layers plus 3 FC layers.

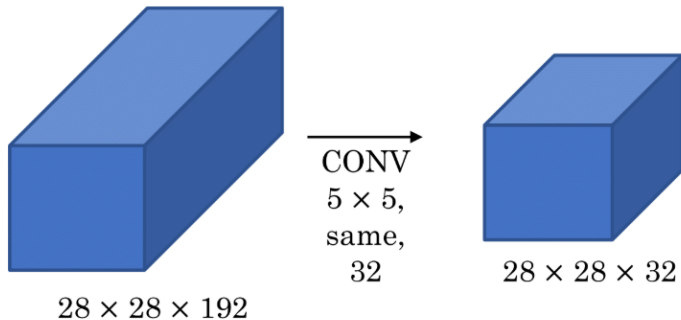


CNN Architectures: **InceptionNet**

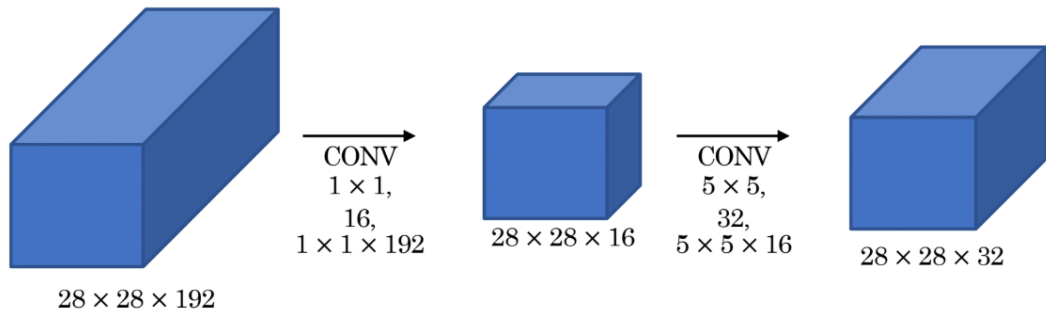
- ▶ InceptionNet goes **deeper** with 22 layers.
- ▶ It is much more **efficient**: only 5 million parameters.
- ▶ Uses a special building block called the **Inception module**.
- ▶ Adds **1x1 convolutions** (bottleneck layers) to make the network faster and smarter.



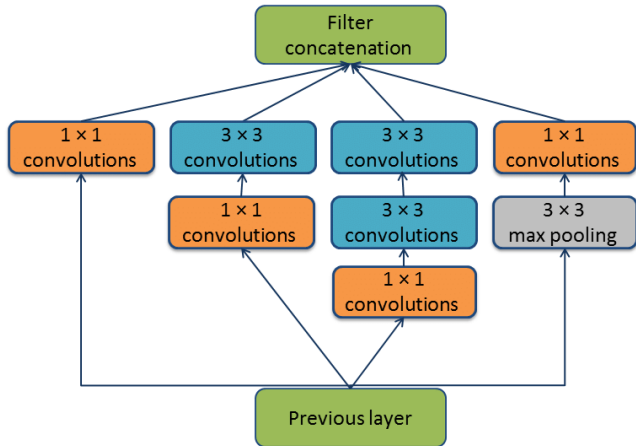
The problem of computational cost

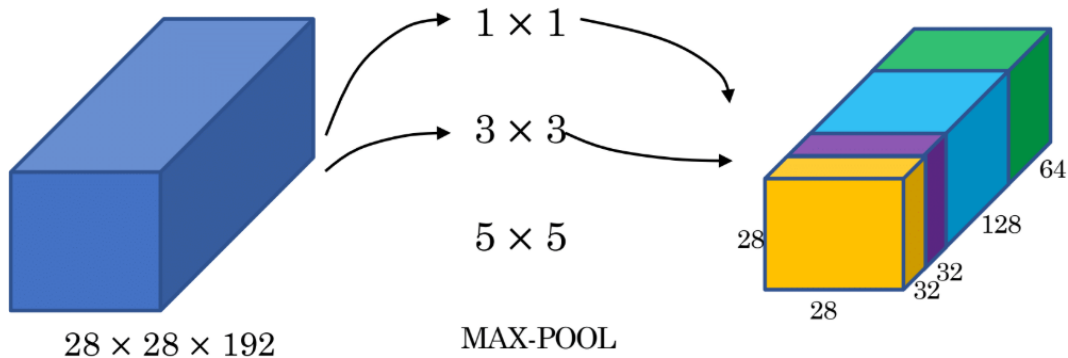


Using 1x1 convolution



- **Inception module:** Uses multiple filter sizes (1×1 , 3×3 , 5×5), in parallel, to capture different features, then combines their outputs.



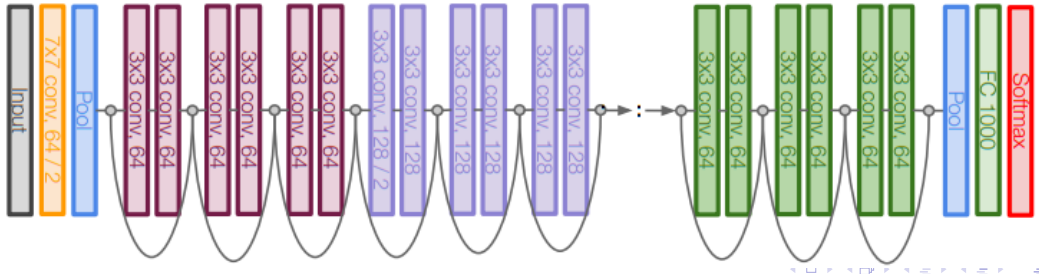


[Szegedy et al. 2014. Going deeper with convolutions]

CNN Architectures: **ResNet**

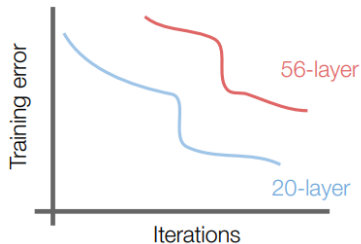
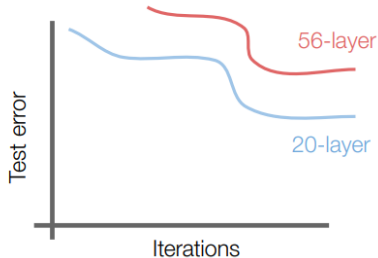
- ▶ **Problem:** Making networks deeper does not always improve accuracy.
- ▶ **Why?** In very deep networks, gradients become extremely small as they move backward through layers, making learning slow or stopping it altogether (**vanishing gradient problem**).
- ▶ **Solution:** Residual Network (ResNet) introduces **skip connections (residuals)**, allowing information to flow more easily.

- ▶ Very deep networks using residual connections
- ▶ 152-layer model for ImageNet
- ▶ Stacked Residual Blocks
- ▶ **Residual:** A shortcut connection that helps the network pass information through layers more easily.

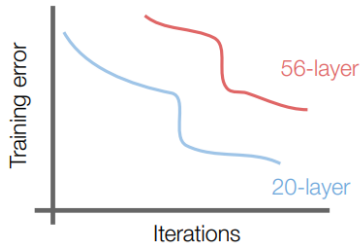
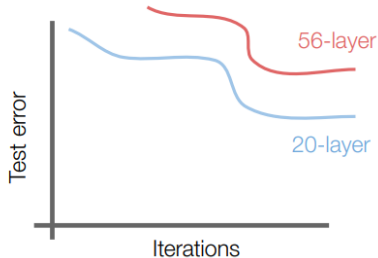


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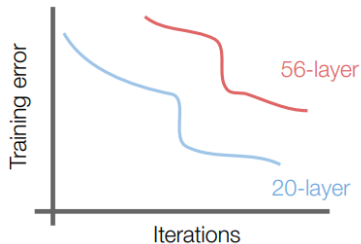
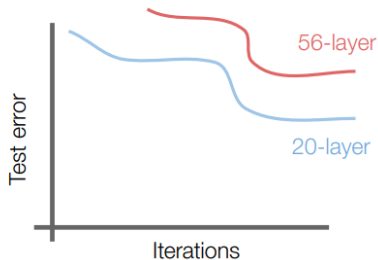


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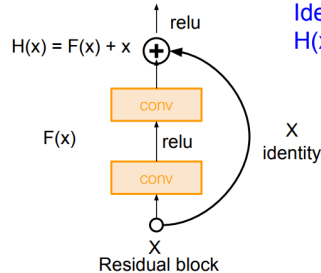
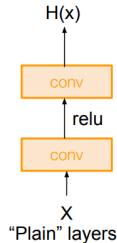
- 56-layer model performs worse on both test and training error
- The deeper model performs worse, but it's not caused by overfitting!

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Identity mapping:
 $H(x) = x$ if $F(x) = 0$

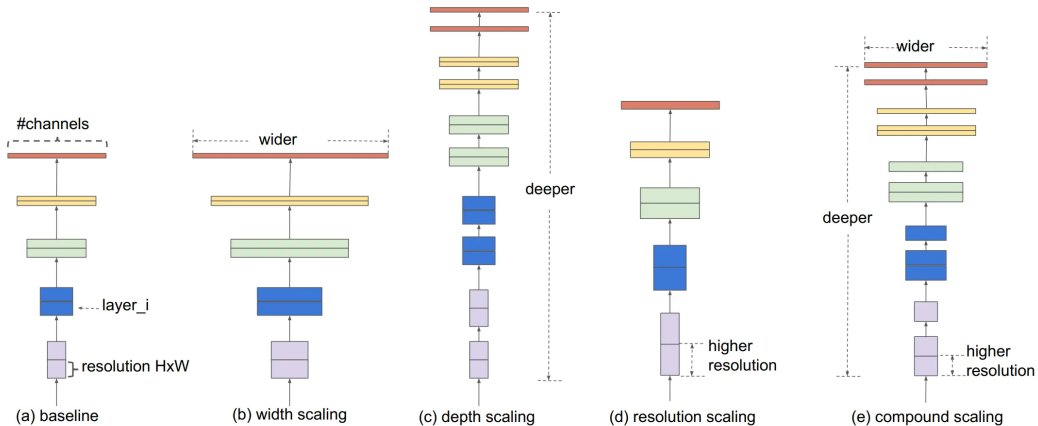
CNN Architectures: **EfficientNet**

► **Problem:** To improve accuracy, we can:

- Increase the number of layers (**depth**)
- Increase the number of neurons in each layer (**width**)
- Use higher-resolution images (**resolution**)

But finding the right balance of these three was largely based on trial and error.

► **Solution:** EfficientNet introduced a **compound scaling** method—a mathematical formula to systematically find the optimal balance among **depth**, **width**, and **resolution**.



EfficientNet Scaling.

- ▶ **Compound Scaling** Uses a single **scaling coefficient** (ϕ) to control:
 - **Network Depth** (α^ϕ) $\rightarrow$ More layers
 - **Network Width** (β^ϕ) $\rightarrow$ More channels per layer
 - **Input Resolution** (γ^ϕ) $\rightarrow$ Larger input images
- ▶ The goal: find α, β, γ that balance accuracy & efficiency, then scale up optimally by increasing the global coefficient ϕ .

- ▶ EfficientNet optimizes depth, width, and resolution using this constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

- ▶ **Why this equation?**

- Increasing **depth** (α) increases FLOPs **linearly**.
 - Increasing **width** (β) increases FLOPs **quadratically** (β^2).
 - Increasing **resolution** (γ) increases FLOPs **quadratically** (γ^2).
- ▶ To double total FLOPs, the three factors must be balanced together.

- ▶ The authors of EfficientNet searched for the best scaling factors on a small baseline model.
- ▶ They found:

$$\alpha = 1.2, \quad \beta = 1.1, \quad \gamma = 1.15$$

- The EfficientNet family (B0 to B7) is generated using:

$$\text{Depth} = 1.2^\phi, \quad \text{Width} = 1.1^\phi, \quad \text{Resolution} = 1.15^\phi$$

Model	ϕ	Depth (α^ϕ)	Width (β^ϕ)
B0	0	$1.2^0 = 1.0$	$1.1^0 = 1.0$
B1	1	$1.2^1 = 1.2$	$1.1^1 = 1.1$
B2	2	$1.2^2 = 1.44$	$1.1^2 = 1.21$
B3	3	$1.2^3 = 1.73$	$1.1^3 = 1.33$
B4	4	$1.2^4 = 2.07$	$1.1^4 = 1.46$
B5	5	$1.2^5 = 2.49$	$1.1^5 = 1.61$
B6	6	$1.2^6 = 2.99$	$1.1^6 = 1.77$
B7	7	$1.2^7 = 3.58$	$1.1^7 = 1.94$

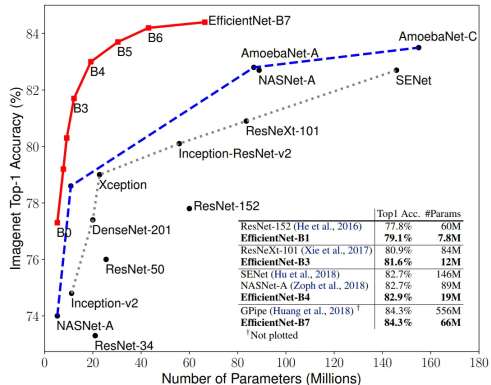
Scaling EfficientNet from B0 to B7

- We multiply these scaling factors by the baseline EfficientNet-B0 values and round to the nearest integer to get the new depth, width, and resolution for each model.

- ▶ EfficientNet models achieve state-of-the-art accuracy with significantly fewer parameters and FLOPs.

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- ▶ EfficientNet-B7 reaches 84.4% Top-1 and 97.3% Top-5 accuracy on ImageNet.
- ▶ More efficient than previous CNN models—8.4x smaller and 6.1x faster than competitors.



EfficientNet Performance on Imagenet.

CNN Architectures: **MobileNet**

► Why MobileNets?

- Small-sized models are crucial for mobile and embedded devices.
- MobileNets reduce computational cost and memory usage while maintaining good accuracy.

► Key Idea:

- Use **depthwise-separable convolutions** to significantly reduce computation compared to standard convolutions.

- **Computational cost of standard convolution:**

$$\text{Cost} = \# \text{ filter params} \times \# \text{ filter positions} \times \# \text{ filters}$$

- Filters operate on all input channels, increasing computation significantly.



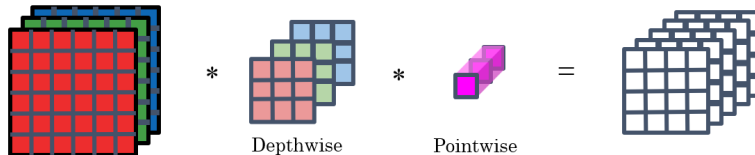
- ▶ Split standard convolution into two steps:
 - **Depthwise Convolution:** Applies a single filter per input channel.
 - **Pointwise Convolution:** Combines outputs from depthwise convolution.
- ▶ **Key Benefit:** Reduces computational cost significantly compared to standard convolution.

Depthwise-Separable Convolutions (cont.)

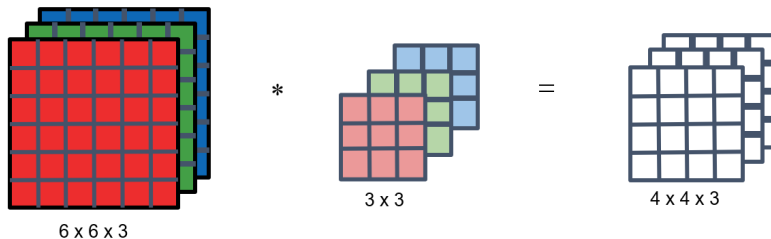
Normal Convolution



Depthwise Separable Convolution

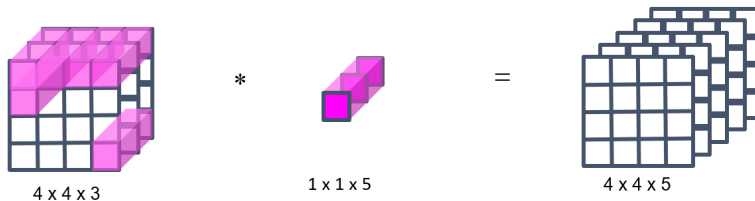


- Operates on each input channel separately.



Computational cost = #filter params x # filter positions x # of filters

- Combines outputs from depthwise convolution using 1×1 convolutions (mixes channels).



$$\text{Computational cost} = \text{\#filter params} \times \text{\# filter positions} \times \text{\# of filters}$$

► MobileNet v2:

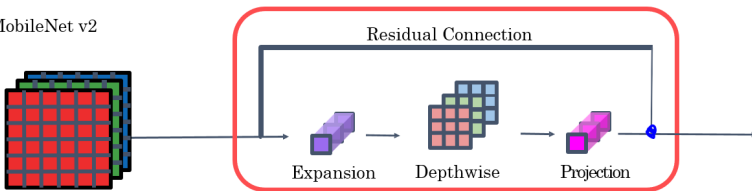
- Adds **residual connections**.
- Introduces:
 - **Expansion step:** Expands input dimensions before depthwise convolution.
 - **Projection step:** Reduces dimensions after processing.

MobileNet Architectures (cont.)

MobileNet v1



MobileNet v2



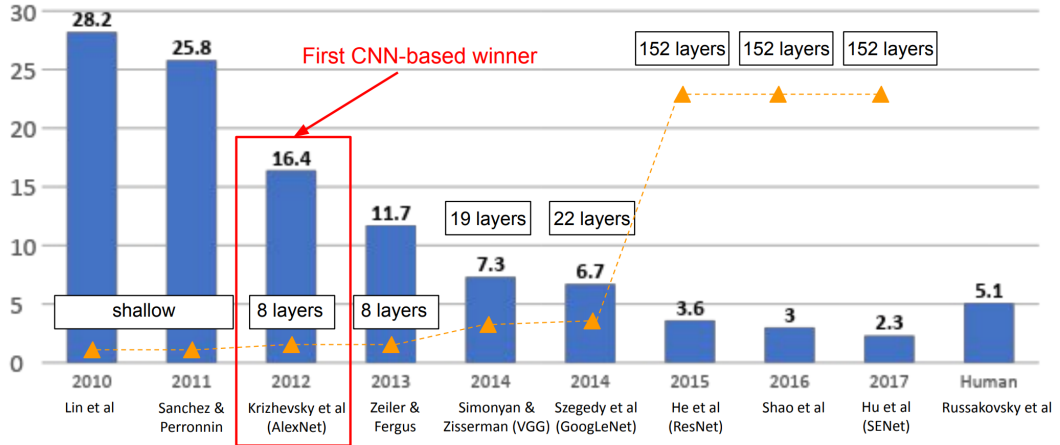
Model	Year	Key Idea	#Params	Strength
LeNet-5	1998	Early CNN for digit recognition	~60K	Pioneered CNNs
AlexNet	2012	Deeper + ReLU + Dropout	~60M	Started Deep Learning
VGG	2014	3×3 filters, uniform design	~138M	Simplicity
Inception	2014	Multi-scale filters	~6.8M	Efficient
ResNet	2015	Skip connections	~25M (50)	Very deep models
EfficientNet	2019	Compound scaling	~5M (B0)	Accuracy/efficiency trade-off
MobileNet	2017	Lightweight architecture	~4M	Mobile-ready

CNN Datasets: **ImageNet**

- ▶ The most extensive data for Image Classification
- ▶ 3 RGB channels from 0 to 255
- ▶ 14,197,122 images
- ▶ 1000 classes



ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



CNN: Loss Functions & Optimizers

Loss:

- ▶ Loss functions measure how well the model's predictions match the true labels.
- ▶ Common loss functions include:
 - Cross-Entropy Loss (for classification)
 - Mean Squared Error (for regression)
- ▶ The choice of loss function depends on the task at hand.

Optimizers:

- ▶ Optimizers update the model's parameters based on the gradients computed during backpropagation.
- ▶ Common optimizers include:
 - Stochastic Gradient Descent (SGD)
 - Adam
 - RMSprop
- ▶ The choice of optimizer can significantly affect the training speed and convergence.

```
import torch.nn as nn
import torch

criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3)
loss = criterion(preds, labels)
loss.backward()
optimizer.step()
```

CNN: Limitations and Future Directions

- ▶ **Understanding global context:** CNNs are inherently local due to convolutional filters, making it hard to capture long-range dependencies.
- ▶ **Variations outside training data:** Performance drops significantly when faced with data distributions not seen during training.
- ▶ **Transfer learning limitations:** Pre-trained CNNs do not always generalize well to new tasks or domains.
- ▶ **BatchNorm sensitivity:** Batch Normalization effectiveness depends on batch size, which can be problematic for small-batch or online learning.
- ▶ **Compute constraints:** Deploying CNNs on mobile or edge devices remains challenging due to high computational and memory requirements.

- ▶ **Vision Transformers (ViTs):** Leveraging self-attention mechanisms to capture global context and long-range dependencies.
- ▶ **Self-supervised learning:** Reducing reliance on labeled data by learning useful representations from unlabeled data.
- ▶ **More efficient models:** Architectures like EfficientNet optimize accuracy and efficiency for deployment on resource-constrained devices.
- ▶ **Better augmentation:** Automated data augmentation techniques (e.g., AutoAugment, RandAugment) improve generalization.
- ▶ **Low-power inference:** Techniques such as quantization and pruning enable CNNs to run efficiently on edge and mobile devices.

CNN: References & Resources

Books:

- ▶ **Deep Learning** by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
- ▶ **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow** by Aurélien Géron
- ▶ **Pattern Recognition and Machine Learning** by Christopher Bishop
- ▶ **Deep Learning for Computer Vision with Python** by Adrian Rosebrock

Online Courses:

- ▶ **Deep Learning Specialization** by Andrew Ng (Coursera)
- ▶ **Convolutional Neural Networks for Visual Recognition** by Fei-Fei Li (Stanford University)
- ▶ **Practical Deep Learning for Coders** by Jeremy Howard (fast.ai)
- ▶ **Deep Learning with PyTorch** by Facebook AI Research

Research Papers:

- ▶ **ImageNet Classification with Deep Convolutional Neural Networks** by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton
- ▶ **Very Deep Convolutional Networks for Large-Scale Image Recognition** by K. Simonyan and A. Zisserman
- ▶ **Deep Residual Learning for Image Recognition** by Kaiming He et al.
- ▶ **Densely Connected Convolutional Networks** by Gao Huang et al.

Websites and Blogs:

- ▶ **Towards Data Science** (Medium)
- ▶ **Distill.pub** (Research and visualization)
- ▶ **Kaggle** (Datasets and competitions)
- ▶ **Papers with Code** (Research papers with code implementations)

YouTube Channels:

- ▶ **3Blue1Brown** (Mathematics and deep learning)
- ▶ **Two Minute Papers** (Research summaries)
- ▶ **Sentdex** (Python programming and machine learning)
- ▶ **StatQuest with Josh Starmer** (Statistics and machine learning)

GitHub Repositories:

- ▶ **TensorFlow Models** (TensorFlow implementations)
- ▶ **PyTorch Examples** (PyTorch implementations)
- ▶ **fastai** (High-level library for deep learning)
- ▶ **Keras** (High-level neural networks API)

Conferences:

- ▶ **NeurIPS** (Neural Information Processing Systems)
- ▶ **CVPR** (Computer Vision and Pattern Recognition)
- ▶ **ICML** (International Conference on Machine Learning)
- ▶ **ICLR** (International Conference on Learning Representations)